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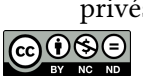
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SHARP STABILITY OF BRENIER MAPS VIA QUANTITATIVE REGULARITY OF POTENTIALS

QUENTIN MÉRIGOT

Abstract. We study the stability of optimal transport maps for the quadratic cost $c(x, y) = \|x - y\|^2$ on $\mathbb{R}^d$ with respect to Wassersein-1 perturbations of the target measure. Existing results establish Hölder stability estimates, but the best-known exponent (1/6) does not match the theoretical lower bound (1/4). In this paper, we close this gap by proving a sharp 1/4-Hölder stability estimate when the source probability distribution has a bounded density. Our approach relies on analyzing the strong convexity of the Kantorovich functional through its associated Bregman-Kantorovich and Jensen-Kantorovich divergences. We show that quantitative bounds on the singularities of convex functions yield meaningful lower bounds on these divergences involving distances between transport maps. Furthermore, these methods also yield quantitative convergence rates for the entropic Brenier map towards the unregularized Brenier map as the regularization parameter vanishes, requiring only mild assumptions on the measures.

1. Introduction

The stability of optimal transport maps has been the focus of significant recent research, driven in part by applications in numerical analysis, statistics, and sampling theory, as well as the need to better understand the "linearized optimal transport" framework. From a PDE perspective, this stability question amounts to quantifying the well-posedness of the (real) Monge-Ampère equation associated to optimal, and to establish continuous dependence on the data even when solutions lacks regularity. In this introduction, we summarize the stability question and existing results in the quadratic case—that is, for the cost $c(x, y) = \|x - y\|^2$ on $\mathbb{R}^d$.

Given a probability density ρ supported on a compact set X of $\mathbb{R}^d$ and a target probability measure μ supported on another compact set Y of $\mathbb{R}^d$, Brenier's theorem guarantees the existence of a unique quadratic optimal transport map, i.e. a unique minimizer in the problem:

$$\inf_{T \in \mathrm{L}^2(\rho, \mathbb{R}^d), T_\# \rho = \mu} \int \|x - T(x)\|^2 \, \mathrm{d}\rho(x).$$

We will call this optimal map the *Brenier map* and we will denote it $T_{\rho \to \mu}$. This map takes the form $T_{\rho \to \mu} = \nabla u$ for a convex potential $u : \mathbb{R}^d \to \mathbb{R}$. While the potential u may not be unique, its gradient is uniquely defined in $\mathrm{L}^2(\rho)$. When ρ satisfies a connectedness assumption, such as a Poincaré inequality, the potential u is then also uniquely determined (ρ-a.e.) up to an additive constant.

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The question of stability focuses on establishing bounds of the form

$$\forall \mu, \nu \in \mathcal{P}(Y), \quad \|T_{\rho\to\mu} - T_{\rho\to\nu}\|_{\mathrm{L}^2(\rho)} \leqslant C\mathcal{W}_p(\mu,\nu)^\alpha, \tag{1.1}$$

where the constants C and α may depend on ρ and Y. In practice, existing stability results are established for $p = 1$, with stability for other exponents deduced from the inequality $\mathcal{W}_1 \leqslant \mathcal{W}_p$. The first stability result of this form is due to Berman [2], who showed that if ρ is bounded away from zero and infinity on a compact set, such inequalities hold with a dimension-dependent α. Delalande and the author [11] later improved to $\alpha = 1/6$, under the same assumptions on ρ, using convex analysis and functional inequalities.

The introduction of a *gluing technique* by Letrouit and the author [19] broadened the scope of the stability theory, enabling its extension to much more general source measures, cost functions, and ambient spaces. In particular, [19] established the inequality (1.1) for source measures ρ whose support lacks convexity, or whose densities tend to 0 or $+\infty$ at controlled rate. We refer to the PhD thesis of Alex Delalande [10] and to the short book of Cyril Letrouit [17] for a detailed account of the existing stability theory, as well as to Remark 1.2. Finally, it is worth noting that this stability literature is rapidly growing, with active research exploring entropic optimal transport [8, 9, 12], other cost functions [22], Riemannian manifolds and more general metric spaces [16, 23], or Schrödinger bridges [25].

1.1. **From stability of potentials to stability of maps.** The general proof strategy to establish stability of optimal transport relies on Kantorovich duality, specifically on the *Kantorovich functional* $\mathcal{K}_\rho : \mathcal{C}^0(Y) \to \mathbb{R}$, defined as

$$\mathcal{K}_\rho(v) = \int v^* \mathrm{d}\rho, \quad \text{where} \quad v^*(x) = \max_{y\in Y} \big[\langle x|y\rangle - v(y)\big].$$

When ρ is a probability density supported on a compact set X, the functional $\mathcal{K}_\rho$ is differentiable everywhere, with gradient

$$\nabla\mathcal{K}_\rho(v) = -\nabla v^*_\# \rho.$$

Constructing the optimal transport map between ρ and μ begins by solving the dual minimization problem:

$$\min_{v\in\mathcal{C}^0(Y)} \mathcal{K}_\rho(v) + \int v \mathrm{d}\mu. \tag{1.2}$$

Denoting v a minimizer to (1.2), the optimality condition for this problem is exactly $\nabla v^*_\# \rho = \mu$, meaning that the Brenier map is given by $T_{\rho\to\mu} = \nabla v^*$. Thus, a natural first step toward establishing the stability of transport maps is to establish the stability of their corresponding dual potentials, solution to (1.2). The convexity of the Kantorovich functional $\mathcal{K}_\rho$ implies the mononicity of its gradient, i.e. $\langle \nabla\mathcal{K}_\rho(v_0) - \nabla\mathcal{K}_\rho(v_1)|v_0 - v_1\rangle \geqslant 0$. Most stability results rely on strenghtened monotonicity inequalities, of the form

$$\min_{c\in\mathbb{R}} \|v_0^* - v_1^* - c\|^2_{\mathrm{L}^2(\rho)} \lesssim \langle \nabla\mathcal{K}_\rho(v_0) - \nabla\mathcal{K}_\rho(v_1)|v_0 - v_1\rangle, \tag{1.3}$$

which are related to *strong* convexity estimates for $\mathcal{K}_\rho$. Here and later, the notation $\lesssim$ hides constants that depend only on X, Y, and ρ. The proof of this strong convexity inequality relies on established geometric inequalities

such as the Brascamp-Lieb or Prékopa-Leindler inequalities. Importantly, such estimates can *only* hold if the potential is unique — note that uniqueness is only up to an additive constant, explaining the minimum over $c \in \mathbb{R}$ on the left-hand side.

In a second step, the stability of optimal transport maps is deduced via a reverse Poincaré inequality bounding the distance between gradients from the distance between *convex* potentials: for any $v_0, v_1 \in \mathcal{C}^0(Y)$,

$$\|\nabla v_0^* - \nabla v_1^*\|_{\mathrm{L}^2(\rho)} \lesssim \min_{c\in\mathbb{R}} \|v_0^* - v_1^* - c\|_{\mathrm{L}^2(\rho)}^{1/3}. \tag{1.4}$$

Combining these estimates yields the final bound. For two target measures $\mu, \nu \in \mathcal{P}(Y)$, we let $v_{\rho\to\mu}, v_{\rho\to\nu} \in \mathcal{C}^0(Y)$ be associated dual potentials, i.e. the maximizers of (1.2). Combining (1.3) and (1.4), the associated Brenier maps $T_{\rho\to\mu} = \nabla v_{\rho\to\mu}^*$ and $T_{\rho\to\nu} = \nabla v_{\rho\to\nu}^*$ then satisfy

$$\begin{aligned}\|T_{\rho\to\mu} - T_{\rho\to\nu}\|_{\mathrm{L}^2(\rho)}^6 &\lesssim \min_{c\in\mathbb{R}} \left\|v_{\rho\to\mu}^* - v_{\rho\to\nu}^* - c\right\|_{\mathrm{L}^2(\rho)}^2 \\ &\lesssim \langle \nu - \mu | v_{\rho\to\mu} - v_{\rho\to\nu}\rangle \lesssim \mathcal{W}_1(\mu,\nu).\end{aligned}$$

While each inequality in this chain is sharp in its exponent, this does not preclude the possibility of achieving a better overall exponent by following an alternative sequence of estimates.

1.2. **Lower bound for the stability exponent.** Gigli [14] provided a counter-example demonstrating that it is not possible to achieve a bound of the form (1.1) for $\alpha > 1/2$ even if ρ is the restriction of the Lebesgue measure to a convex set. More recently, Letrouit [18, Theorem 1.2] proved that for dimensions $d \geqslant 2$, there exist a bounded open set $X \subset \mathbb{R}^d$ and a compact set Y such that, if ρ denotes the uniform probability measure on X, there exists a bounded and compactly supported source measure ρ such that the stability inequality (1.1) cannot hold for any exponent $\alpha > p/(2p+2)$. In particular, for $p = 1$, the exponent in (1.1) cannot be larger than $\frac{1}{4}$ for a general bounded and compactly supported source measure ρ.

1.3. **Contributions.** To quantify the strong convexity of the Kantorovich functional $\mathcal{K}_\rho$, we introduce its associated Bregman and Jensen divergences in §2. By leveraging recent quantitative results on the singularities of convex functions [4] alongside elementary arguments from convex analysis, we establish tight lower bounds on these divergences, see Theorem 2.2. In Appendix A, we provide a second proof of Theorem 2.2.(ii) with slightly different techniques, which are subsequently applied in §4.

From these estimates, we deduce a proof of the stability inequality (1.1) in §3 for $p = 1$ with the optimal exponent $\alpha = 1/4$ for bounded densities ρ.

THEOREM (Theorem 3.1). *Let X, Y be compact and convex subsets of $\mathbb{R}^d$, let $\rho \in \mathcal{P}^{\mathrm{ac}}(X) \cap \mathrm{L}^\infty(X)$. Then* (1.1) *holds for $p = 1$ and $\alpha = \frac{1}{4}$.*

Remark 1.1 (Optimal exponent and mildness of hypotheses). Theorem 3.1 substantially improves upon recent stability results for optimal transport maps, such as [11, Theorem 4.2] and [19, Theorem 1.7], in two ways. First, the Hölder exponent of 1/4 improves upon the exponent of 1/6 obtained in those works, and by the counterexample constructed by Letrouit [18] is

is optimal for compactly supported measures with bounded densities. Second, our hypotheses are also significantly milder: we only require an upper bound on the source density, completely bypassing the need for uniform lower bounds on ρ or connectedness assumptions on the support (such as being a John domain) — even if finite unions of connected sets may also be treated with these techniques.

Remark 1.2 (Map stability versus potential stability). While the present work provides an optimal stability theory for Brenier *maps*, it does not imply an optimal stability theory for the associated convex *potentials*. For instance, if one assumes that ρ satisfies a Poincaré inequality, bounding the variance between the Brenier potentials using the $\mathrm{L}^2(\rho)$ map stability of Theorem 3.1 yields

$$\min_{c\in\mathbb{R}} \|u_{\rho\to\mu} - u_{\rho\to\nu} - c\| \lesssim \mathcal{W}_1(\mu,\nu)^{1/4}. \tag{1.5}$$

This 1/4 exponent is suboptimal for potentials, while the functional functional inequality approach recover the optimal 1/2 exponent (see (1.3)). Furthermore, the gluing techniques of [19] allow one to "localize" the stability problem, yielding results for general costs and spaces [16, 15] and optimal transport variants [25]. In hindsight, the stability theory started in [11, 19] is perfectly tailored to the stability of potentials (and may possibly be applied even when *no* optimal transport map exists, such as for non-twisted costs), whereas the variational approach developed in this article recovers sharp stability of maps in the Brenier setting, i.e. quadratic cost over $\mathbb{R}^d$.

Finally, in §4 we apply these techniques to quantify the convergence of the *entropic Brenier map* T_ε, i.e. the barycentric projection of the entropy-regularized optimal plan γ_ε, toward the unregularized Brenier map as the regularization parameter goes to zero, requiring only very mild regularity assumptions. The upper bound involves the optimality gap

$$P_\varepsilon = \int \langle x|T_0(x)\rangle \mathrm{d}\rho(x) - \int \langle x|y\rangle \mathrm{d}\gamma_\varepsilon(x,y) \geqslant 0,$$

whose estimation has been the object of many recent work in the optimal transport literature. We refer to §4 for the definition of γ_ε and T_ε.

Theorem (Theorem 4.3). *Let $X, Y \subseteq \mathbb{R}^d$ be compact and convex. Assume that $\rho \in \mathcal{P}^{\mathrm{ac}}(X)$ has bounded density and that $\mu \in \mathcal{P}(Y)$. Then,*

$$\|T_\varepsilon - T_0\|^2_{\mathrm{L}^2(\rho)} \leqslant C\sqrt{P_\varepsilon}$$

While finishing this article, we learned of a very recent preprint by Cazelles, Pauwels and Portales [6]. Among other results, the authors also obtain improvements on the exponent in the stability inequality (1.1) using the quantitative estimates on the singularities of convex functions proven in [4], and a theorem similar to Theorem 3.4. However, the motivation and scope of their contribution differs from ours, and their precise proof strategy do not yield the optimal exponent $\alpha = \frac{1}{4}$ in (1.1).

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2. The Jensen-Kantorovich and Bregman-Kantorovich divergences

In this section, we introduce the Bregman and Jensen divergences associated with the Kantorovich functional and establish our main technical contribution : a quantitative lower bounds on these divergences in terms of the $\mathrm{L}^2(\rho)$-distance between optimal transport maps (Theorem 2.2).

Definition 2.1 (Kantorovich functional)**.** Let X and Y be compact convex subsets of $\mathbb{R}^d$ and let ρ be a probability density on X. We define the *Kantorovich functional* $\mathcal{K}_\rho : \mathcal{C}^0(Y) \to \mathbb{R}$ by

$$\mathcal{K}_\rho(v) = \int_X v^*(x) \mathrm{d}\rho(x),$$

where $v^*(x) = \max_{y\in Y} \langle x|y\rangle - v(y)$ is the convex conjugate of the function v extended by $+\infty$ outside of Y.

The gradient of the Kantorovich functional $\mathcal{K}_\rho$ will be taken with respect to the duality $\mathcal{C}^0(Y)^* = \mathcal{M}(Y)$. A simple computation, which may for instance be found in [11, Proposition 2.2], shows that $\nabla\mathcal{K}_\rho(v) = -\nabla v^* \# \rho$.

Definition 2.2 (Divergences)**.** We define the Jensen-Kantorovich (J_ρ), Bregman-Kantorovich (B_ρ) and symmetrized Bregman-Kantorovich (S_ρ) divergence between two potentials $v_0, v_1 \in \mathcal{C}^0(Y)$ by

$$J_\rho(v_0, v_1) = \frac{1}{2}\mathcal{K}_\rho(v_0) + \frac{1}{2}\mathcal{K}_\rho(v_1) - \mathcal{K}_\rho\left(\frac{1}{2}(v_0 + v_1)\right), \tag{2.6}$$

$$B_\rho(v_0, v_1) = \mathcal{K}_\rho(v_1) - \mathcal{K}_\rho(v_0) - \langle \nabla\mathcal{K}_\rho(v_0)|v_1 - v_0\rangle, \tag{2.7}$$

$$S_\rho(v_0, v_1) = \langle \nabla\mathcal{K}_\rho(v_1) - \nabla\mathcal{K}_\rho(v_0)|v_1 - v_0\rangle. \tag{2.8}$$

Since the Kantorovich functional $\mathcal{K}_\rho$ is convex, these divergences are non-negative. They are related in the following way.

Proposition 2.1. *For all $v_0, v_1 \in \mathcal{C}^0(Y)$,*

$$J_\rho(v_0, v_1) \leqslant \frac{1}{2} B_\rho(v_0, v_1) \tag{2.9}$$

$$S_\rho(v_0, v_1) = B_\rho(v_0, v_1) + B_\rho(v_1, v_0). \tag{2.10}$$

Proof. From the convexity of the Kantorovich functional $\mathcal{K}_\rho$, we obtain

$$\mathcal{K}_\rho\left(\frac{v_0 + v_1}{2}\right) \geqslant \mathcal{K}_\rho(v_0) + \langle \nabla\mathcal{K}_\rho(v_0)|\frac{v_1 - v_0}{2}\rangle.$$

Substituting this lower bound into the definition of the J_ρ yields:

$$\begin{aligned} J_\rho(v_0, v_1) &= \frac{1}{2}\mathcal{K}_\rho(v_0) + \frac{1}{2}\mathcal{K}_\rho(v_1) - \mathcal{K}_\rho\left(\frac{v_0+v_1}{2}\right) \\ &\leqslant \frac{1}{2}\mathcal{K}_\rho(v_0) + \frac{1}{2}\mathcal{K}_\rho(v_1) - \left[\mathcal{K}_\rho(v_0) + \frac{1}{2}\langle \nabla\mathcal{K}_\rho(v_0) | v_1 - v_0\rangle\right] \\ &= \frac{1}{2}\left[\mathcal{K}_\rho(v_1) - \mathcal{K}_\rho(v_0) - \langle \nabla\mathcal{K}_\rho(v_0) | v_1 - v_0\rangle\right] = \frac{1}{2}B_\rho(v_0, v_1). \quad \square \end{aligned}$$

The next theorem is the main theorem of this article. It establishes lower bounds on the divergences associated to the Kantorovich functional using the $\mathrm{L}^2(\rho)$-distance between the optimal transport maps ∇v_0^* and ∇v_1^*. These lower bounds can be regarded as strong convexity estimates for $\mathcal{K}_\rho$.

THEOREM 2.2. *Let X and Y be compact and convex subsets of $\mathbb{R}^d$, and let ρ be a probability density on X with density bounded by M_ρ. Then, for all $v_0, v_1 \in \mathcal{C}^0(Y)$,*

(i) $\|\nabla v_0^* - \nabla v_1^*\|^4_{\mathrm{L}^2(\rho)} \leqslant C J_\rho(v_0, v_1)$.
(ii) $\|\nabla v_0^* - \nabla v_1^*\|^4_{\mathrm{L}^2(\rho)} \leqslant \frac{C}{2} B_\rho(v_0, v_1)$
(iii) $\|\nabla v_0^* - \nabla v_1^*\|^4_{\mathrm{L}^2(\rho)} \leqslant \frac{C}{4} S_\rho(v_0, v_1)$

where $C = 2^8 c_d M_\rho \operatorname{diam}(Y)^3 \operatorname{diam}(X)^{d-1}$ and c_d is the constant from Theorem 2.3.

The proof of this theorem relies on a theorem quantifying the singularities of convex functions, stated as Corollary 2.2 in [4].

THEOREM 2.3 (Carlier, Delalande, M.). *Let $w : \mathbb{R}^d \to \mathbb{R}$ be convex and Lipschitz. Then one has for any $R > 0$ and any $\eta \geqslant 0$,*

$$\int_{\mathrm{B}(0,R)} \operatorname{diam}\left(\partial w(\mathrm{B}(x,\eta))\right)^2 \mathrm{d}x \leqslant c_d (R+\eta)^{d-1} \operatorname{Lip}(w)^2 \eta,$$

where $c_d = 48 d^2 \beta_d 2^{3d+2}$ and β_d is the volume of the unit d-ball.

We will prove the lower bound on the Jensen divergence J_ρ. The inequalities (ii) and (iii) from Theorem 2.2 then directly follow from Proposition 2.1. The proof of Theorem 2.2.(i) uses the following representation formula for the Jensen-Kantorovich divergence J_ρ.

LEMMA 2.4. *Given $v_0, v_1 \in \mathcal{C}^0(Y)$, the Jensen divergence admits the following representation, where $u_i = v_i^*$:*

$$J_\rho(v_0, v_1) = \int_X \sup_{h \in \mathbb{R}^d} \left\{ \frac{u_0(x) + u_1(x) - u_0(x+h) - u_1(x-h)}{2} \right\} \mathrm{d}\rho(x). \tag{2.11}$$

Proof. We first note that by definition,

$$J_\rho(v_0, v_1) = \int \frac{1}{2}(v_0^* + v_1^*) - \left(\frac{v_0+v_1}{2}\right)^* \mathrm{d}\rho.$$

The convex conjugate of a sum is an inf-convolution, thus implying that

$$\left(\frac{v_0+v_1}{2}\right)^*(x) = \inf_{y_0+y_1=2x} \left(\frac{1}{2}u_0(y_0) + \frac{1}{2}u_1(y_1)\right).$$

Changing variables to $y_0 = x + h$ and $y_1 = x - h$ yields the result. □

Proof of Theorem 2.2.(i). Since the divergence J_ρ and the $\mathrm{L}^2(\rho)$-distance between gradients are invariant under translations of the source and target domains, we may assume without loss of generality that the origin 0 belongs to both X and Y. We denote their respective radii from the origin by

$$\mathcal{R}_X = \max_{x\in X} \|x\| \leqslant \operatorname{diam}(X), \qquad \mathcal{R}_Y = \max_{y\in Y} \|y\| \leqslant \operatorname{diam}(Y). \tag{2.12}$$

Consequently, for any $v \in \mathcal{C}^0(Y)$, the supremum in the definition of the convex conjugate v^* is taken over $Y \subseteq \mathrm{B}(0, \mathcal{R}_Y)$, which immediately implies that v^* is globally $\mathcal{R}_Y$-Lipschitz. Let $u_i = v_i^*$ and denote $j(x)$ the integrand in the representation formula (2.11). To get a lower bound on $j(x)$, we will evaluate at a specific, suitably chosen $h(x)$. We have:

$$j(x) \geqslant \frac{1}{2}(u_0(x) - u_0(x + h(x))) + \frac{1}{2}(u_1(x) - u_1(x - h(x))).$$

By the subgradient inequality for convex functions,

$$u_0(x) - u_0(x + h(x)) \geqslant \langle \nabla u_0(x + h(x))| - h(x)\rangle,$$
$$u_1(x) - u_1(x - h(x)) \geqslant \langle \nabla u_1(x - h(x))|h(x)\rangle.$$

Summing these gives:

$$j(x) \geqslant \frac{1}{2}\langle \nabla u_1(x - h(x)) - \nabla u_0(x + h(x))|h(x)\rangle.$$

We now choose $h(x) = \lambda(\nabla u_1(x) - \nabla u_0(x))$ for some $\lambda > 0$. Since $\nabla u_1(x)$ and $\nabla u_0(x)$ both belong to the convex set Y, we note that $\|h(x)\| \leqslant 2R_Y$. This gives us :

$$j(x) \geqslant \frac{\lambda}{2}\langle \nabla u_1(x - h(x)) - \nabla u_0(x + h(x))|\nabla u_1(x) - \nabla u_0(x)\rangle.$$

We decompose the left side of the inner product as:

$$\begin{aligned}\nabla u_1(x - h) - \nabla u_0(x + h) = (\nabla u_1(x) - \nabla u_0(x)) + (\nabla u_1(x - h) - \nabla u_1(x)) \\ - (\nabla u_0(x + h) - \nabla u_0(x)).\end{aligned}$$

Injecting this decomposition, we have

$$\begin{aligned}j(x) \geqslant \frac{\lambda}{2}\|\nabla u_1(x) - \nabla u_0(x)\|^2 + \frac{\lambda}{2}\langle \nabla u_1(x - h) - \nabla u_1(x)|\nabla u_1(x) - \nabla u_0(x)\rangle \\ - \frac{\lambda}{2}\langle \nabla u_0(x + h) - \nabla u_0(x)|\nabla u_1(x) - \nabla u_0(x)\rangle.\end{aligned}$$

Using the inequality $|\langle a|b\rangle| \leqslant \frac{1}{4}\|b\|^2 + \|a\|^2$ on both error terms, we absorb a factor of $\frac{1}{8}$ into the main term twice, leaving:

$$\begin{aligned}j(x) \geqslant \frac{\lambda}{4}\|\nabla u_1(x) - \nabla u_0(x)\|^2 - \frac{\lambda}{2}\|\nabla u_1(x - h) - \nabla u_1(x)\|^2 \\ - \frac{\lambda}{2}\|\nabla u_0(x + h) - \nabla u_0(x)\|^2.\end{aligned}$$

We now integrate over X. Since $\|h(x)\| \leqslant \eta_\lambda := 2\mathcal{R}_Y\lambda$, the error terms are bounded by the diameter of the subdifferentials of u_i on balls of radius η_λ:

$$\int_X \|\nabla u_i(x \pm h(x)) - \nabla u_i(x)\|^2 \,\mathrm{d}\rho(x) \leqslant M_\rho \int_X \operatorname{diam}(\partial u_i(B(x, \eta_\lambda)))^2 \mathrm{d}x$$

Applying Theorem 2.3 with $\eta = \eta_\lambda$, we obtain for $i \in \{0, 1\}$,

$$\int_X \|\nabla u_i(x \pm h(x)) - \nabla u_i(x)\|^2 \, \mathrm{d}\rho(x) \leqslant \lambda C_\lambda,$$

where $C_\lambda = C_0(1 + 2(\mathcal{R}_Y/\mathcal{R}_X) \cdot \lambda)^{d-1}$ and $C_0 = 2c_d M_\rho \mathcal{R}_Y^3 \mathcal{R}_X^{d-1}$. Setting $A = \|\nabla u_1 - \nabla u_0\|^2_{\mathrm{L}^2(\rho)}$, we therefore have for all $\lambda \geqslant 0$,

$$J_\rho(v_0, v_1) \geqslant \frac{\lambda}{4} A - \lambda^2 C_\lambda. \tag{2.13}$$

To find an upper bound, we let $\overline{\lambda} = A/(8C_0)$. Since $\nabla u_i(x) \in Y \subseteq \mathrm{B}(0, \mathcal{R}_Y)$, we have the trivial bound $A \leqslant 4\mathcal{R}_Y^2$. In addition, since ρ is a probability density supported on $X \subseteq \mathrm{B}(0, \mathcal{R}_X)$ and bounded from above M_ρ, its total mass satisfies $1 = \rho(X) \leqslant M_\rho \beta_d \mathcal{R}_X^d$, where β_d is the volume of the unit ball in $\mathbb{R}^d$. Combining both inequalities gives us

$$\overline{\lambda} = \frac{A}{8C_0} \leqslant \frac{4\mathcal{R}_Y^2}{8(2c_d/\beta_d)\mathcal{R}_Y^3/\mathcal{R}_X} = \frac{\beta_d}{4c_d} \frac{\mathcal{R}_X}{\mathcal{R}_Y}.$$

Using this upper bound and the explicit value $c_d = 48d^2\beta_d 2^{3d+2}$ from Theorem 2.3, we can bound $C_{\overline{\lambda}}$:

$$C_{\overline{\lambda}} = C_0 \left(1 + \frac{\beta_d}{2c_d}\right)^{d-1} \leqslant \frac{3}{2} C_0.$$

Substituting $\overline{\lambda}$ into (2.13) gives the desired conclusion with $C = 128C_0$:

$$J_\rho(v_0, v_1) \geqslant \frac{\overline{\lambda}}{4} A - \overline{\lambda}^2 C_{\overline{\lambda}} \geqslant \frac{1}{128 C_0} \|\nabla u_1 - \nabla u_0\|^4_{\mathrm{L}^2(\rho)}. \qquad \square$$

3. Applications to stability of optimal transport

In this section we will see how the lower bound on the Jensen divergence provided in Theorem 2.2 leads to optimal stability results for the optimal transport maps. In this section, we will still assume that X, Y are compact convex subsets of $\mathbb{R}^d$. The assumption that X (and similarly Y) is convex is mild, as we merely require the support of the source measure ρ to be *included* in X, because the constants in our inequalities only depend on the diameter of X. Given $\mu \in \mathcal{P}(X)$, we recall that the Brenier transport map $T_{\rho\to\mu}$ is of the form $T_{\rho\to\mu} = \nabla v^*_{\rho\to\mu}$ where the (dual) potential $v_{\rho\to\mu}$ is a minimizer of

$$v \in \mathcal{C}^0(Y) \mapsto \mathcal{K}_{\rho,\mu}(v) := \mathcal{K}_\rho(v) + \int v \mathrm{d}\mu = \int v^* \mathrm{d}\rho + \int v \mathrm{d}\mu.$$

Without loss of generality, we will often make the assumption that

$$v_{\rho\to\mu}(y) = \max_{x \in X} \langle x | y \rangle - v^*_{\rho\to\mu}(x). \tag{3.14}$$

This assumption then implies that for almost every point y, $\nabla v_{\rho\to\mu}(y)$ belongs to X, leading to Lipschitz estimates on the potential.

3.1. **Stability of optimal transport maps.** Our first application is a proof ofHölder stability of the optimal transport map with respect to the target measure in the Wasserstein distance with the optimal exponent $\frac{1}{4}$.

THEOREM 3.1. *Let X, Y be compact convex subsets of $\mathbb{R}^d$, let ρ be a probability density on X with bounded density. Then,*

$$\forall \mu, \nu \in \mathcal{P}(Y), \quad \|T_{\rho\to\mu} - T_{\rho\to\nu}\|^4_{\mathrm{L}^2(\rho)} \leqslant C' \mathcal{W}_1(\mu, \nu),$$

where $C' = \mathrm{diam}(Y)C/4$ and C is the constant from Theorem 2.2.

Proof. Let $v_{\rho\to\mu}, v_{\rho\to\nu} \in \mathcal{C}^0(Y)$ be the dual Kantorovich potentials associated with μ and ν, and assume that (3.14) holds for both potentials. This implies that $\nabla v_{\rho\to\mu}$ and $\nabla v_{\rho\to\nu}$ both belong to X almost-everywhere, implying that $v_{\rho\to\mu} - v_{\rho\to\nu}$ is $\mathrm{diam}(X)$-Lipschitz. Then, by definition of the symmetrized Bregman-Kantorovich divergence we have

$$S_\rho(v_\nu, v_\mu) = \langle \nabla \mathcal{K}_\rho(v_\mu) - \nabla \mathcal{K}_\rho(v_\nu) | v_\mu - v_\nu \rangle = \int_Y (v_\mu - v_\nu) \mathrm{d}(\nu - \mu).$$

Combining Theorem 2.2 and Kantorovich-Rubinstein duality, we obtain

$$\begin{aligned} \left\| \nabla v^*_\mu - \nabla v^*_\nu \right\|^4_{\mathrm{L}^2(\rho)} &\leqslant \frac{C}{4} S(v_\nu, v_\mu) = \frac{C}{4} \int_Y (v_\mu - v_\nu) \mathrm{d}(\nu - \mu) \\ &\leqslant \frac{C}{4} \mathrm{diam}(Y) \mathcal{W}_1(\mu, \nu). \end{aligned}$$

□

3.2. **Stability with respect to suboptimality.** We can use Theorem 2.2 to bound the distance between the gradient of a suboptimal potential and the true optimal transport map from the "optimality gap", which measures how far the other potential is from optimality.

THEOREM 3.2. *Let X, Y be compact subsets of $\mathbb{R}^d$, let ρ be a probability density on X with bounded density. Let $\mu \in \mathcal{P}(Y)$ and let $T_{\rho\to\mu}$ be the Brenier map. Then,*

$$\forall v \in \mathcal{C}^0(Y), \quad \|\nabla v^* - T_{\rho\to\mu}\|^4_{\mathrm{L}^2(\rho)} \leqslant \frac{C}{2} \left(\mathcal{K}_{\rho,\mu}(v) - \min_{\mathcal{C}^0(Y)} \mathcal{K}_{\rho,\mu} \right),$$

where C is the constant from Theorem 2.2.

Proof. Because $v_0 := v_{\rho\to\mu}$ is the minimizer of $\mathcal{K}_{\rho,\mu}$, evaluating this functional at the midpoint $v_m = \frac{v_0+v}{2}$ yields $\mathcal{K}_{\rho,\mu}(v_m) \geqslant \mathcal{K}_{\rho,\mu}(v_0)$. Rearranging this inequality provides a lower bound on the Kantorovich functional at v_m:

$$-\mathcal{K}_\rho(v_m) \leqslant -\mathcal{K}_\rho(v_0) - \frac{1}{2} \int_Y (v_0 - v) \mathrm{d}\mu.$$

Plugging this inequality into the definition of the Jensen-Kantorovich divergence gives an upper bound:

$$\begin{aligned} J_\rho(v_0, v) &\leqslant \frac{1}{2} \mathcal{K}_\rho(v_0) + \frac{1}{2} \mathcal{K}_\rho(v) - \mathcal{K}_\rho(v_0) - \frac{1}{2} \int_Y (v_0 - v) \mathrm{d}\mu \\ &= \frac{1}{2} \left[\left(\mathcal{K}_\rho(v) + \int_Y v \mathrm{d}\mu \right) - \left(\mathcal{K}_\rho(v_0) + \int_Y v_0 \mathrm{d}\mu \right) \right]. \end{aligned}$$

Applying Theorem 2.2.(i) to lower bound $J_\rho(v_0, v)$ yields the result. □

3.3. **Stability under source and target perturbations.** We can leverage the stability with respect to suboptimality established in Theorem 3.2 to understand the stability of transport maps when both the source and target measures are simultaneously perturbed.

THEOREM 3.3. *Let $X, Y \subseteq \mathbb{R}^d$ be compact subsets and let $\rho_0, \rho_1 \in \mathcal{P}(X)$ and $\mu_0, \mu_1 \in \mathcal{P}(Y)$. Assume that ρ_0 has a bounded density on X. Let $v_{\rho_i \to \mu_i} \in \arg\min_{\mathcal{C}^0(Y)} \mathcal{K}_{\rho_i,\mu_i}$ be an optimal (dual) potential, and let $T_{\rho_i \to \mu_i} = \nabla v^*_{\rho_i \to \mu_i}$. Then,*

$$\|T_{\rho_0 \to \mu_0} - T_{\rho_1 \to \mu_1}\|^4_{\mathrm{L}^2(\rho_0)} \leqslant \frac{C}{2} \left(2 \operatorname{diam}(Y) \mathcal{W}_1(\rho_0, \rho_1) + 2 \operatorname{diam}(X) \mathcal{W}_1(\mu_0, \mu_1)\right),$$

where C is the constant in Theorem 2.2 (associated with $M_{\rho_0} = \|\rho_0\|_{\mathrm{L}^\infty(Y)}$).

Proof. Define $v_i = v_{\rho_i \to \mu_i}$ and $T_i = \nabla v_i^*$. Choosing v_{1-i} as a candidate potential for the problem of transporting ρ_i to μ_i, we obtain

$$\Delta_i := \mathcal{K}_{\rho_i,\mu_i}(v_{1-i}) - \mathcal{K}_{\rho_i,\mu_i}(v_i) \geqslant 0.$$

Thus,

$$\begin{aligned}
\Delta_0 &= \mathcal{K}_{\rho_0,\mu_0}(v_1) - \mathcal{K}_{\rho_0,\mu_0}(v_0) \\
&\leqslant \mathcal{K}_{\rho_0,\mu_0}(v_1) - \mathcal{K}_{\rho_0,\mu_0}(v_0) + \mathcal{K}_{\rho_1,\mu_1}(v_0) - \mathcal{K}_{\rho_1,\mu_1}(v_1) \\
&= \langle \rho_0 - \rho_1 | v_1^* - v_0^* \rangle + \langle \mu_0 - \mu_1 | v_1 - v_0 \rangle
\end{aligned}$$

Since ∇v_i^* belongs to Y, the function $v_1^* - v_0^*$ is $\operatorname{diam}(Y)$-Lipschitz. Assuming that v_i satisfy (3.14), we get as before $\operatorname{Lip}(v_1 - v_0) \leqslant \operatorname{diam}(X)$. Applying Kantorovich-Rubinstein duality gives us a bound on the suboptimality:

$$\Delta_0 \leqslant \operatorname{diam}(Y) \mathcal{W}_1(\rho_0, \rho_1) + \operatorname{diam}(X) \mathcal{W}_1(\mu_0, \mu_1).$$

We conclude by applying Theorem 3.2. □

3.4. **Stability of suboptimal plans.** Variants of the techniques presented before can be used to prove that any near-optimal transport *plan* $\gamma \in \Gamma(\rho, \mu)$ has to be quantitatively close to the graph of the Brenier map $T_{\rho \to \mu}$.

THEOREM 3.4. *Let $X, Y \subseteq \mathbb{R}^d$ be compact and convex. Assume that $\rho \in \mathcal{P}^{\mathrm{ac}}(X)$ has bounded density and let $\mu \in \mathcal{P}(Y)$. Denote $T_0 = T_{\rho \to \mu}$ the Brenier map from ρ to μ, and let $\gamma \in \Gamma(\rho, \mu)$. Then,*

$$\int_{X \times Y} \|y - T_0(x)\|^2 \, \mathrm{d}\gamma(x, y) \leqslant C \left(\int_X \langle x | T_0(x) \rangle \mathrm{d}\rho(x) - \int_{X \times Y} \langle x | y \rangle \mathrm{d}\gamma(x, y) \right)^{1/2}, \tag{3.15}$$

where the constant C depends only on $M_\rho, \mathcal{R}_X, \mathcal{R}_Y$, and d.

Remark 3.1. Unlike Theorems 3.1, 3.2, and 3.3, the next theorem does not follow directly from our lower bounds on the Bregman-Kantorovich or Jensen-Kantorovich divergences. However, its proof is inspired by our initial, direct proof of the Bregman-Kantorovich lower bound, which we have included in Appendix A for completeness.

Proof. Let $v_0 \in \mathcal{C}^0(Y)$ be convex such that $T_0 = \nabla v_0^*$. To bound the error in (3.15), we use the proximal mapping. We refer to Appendix A for its

definition. For any $\lambda > 0$, we define $x_\lambda(x,y) := \mathrm{prox}_{\lambda v_0^*}(x + \lambda y)$. The first-order optimality condition for the proximal operator gives

$$\frac{x_\lambda - x}{\lambda} - y = -\nabla v_0^*(x_\lambda).$$

We decompose the squared distance using the triangle inequality:

$$\begin{aligned}\|y - \nabla v_0^*(x)\|^2 &\leqslant 2\,\|y - \nabla v_0^*(x_\lambda)\|^2 + 2\,\|\nabla v_0^*(x_\lambda) - \nabla v_0^*(x)\|^2 \\ &\leqslant \frac{2}{\lambda^2}\,\|x_\lambda - x\|^2 + 2\,\|\nabla v_0^*(x_\lambda) - \nabla v_0^*(x)\|^2\,.\end{aligned}$$

To bound the first term, we use Lemma A.2, which gives a lower bound on $\|x - x_\lambda\|$ in term of the Fenchel duality gap $G_{v_0^*}(x,y) := v_0^*(x) + v_0(y) - \langle x|y\rangle$

$$\|x - x_\lambda\|^2 \leqslant \lambda G_{v_0^*}(x,y).$$

To bound the second term, we observe that $x_\lambda - x = \lambda(y - \nabla v_0^*(x_\lambda))$. Since the points y and $\nabla v_0^*(x_\lambda)$ both belong to Y, their distance is bounded by $\mathrm{diam}(Y)$, ensuring that $\|x_\lambda - x\| \leqslant \eta_\lambda := \lambda\,\mathrm{diam}(Y)$. Integrating these inequalities over γ, we obtain

$$\begin{aligned}\int_{X\times Y} \|y - \nabla v_0^*(x)\|^2\,\mathrm{d}\gamma(x,y) \leqslant \frac{2}{\lambda}\int_{X\times Y} G_{v_0^*}(x,y)\mathrm{d}\gamma(x,y) \\ + 2\int_X \mathrm{diam}(\partial v_0^*(\mathrm{B}(x,\eta_\lambda)))^2\mathrm{d}\rho(x).\end{aligned}$$

Using Theorem 2.3, we have

$$\int_X \mathrm{diam}(\partial v_0^*(\mathrm{B}(x,\eta_\lambda)))^2\mathrm{d}\rho(x) \leqslant c_d M_\rho(\mathcal{R}_X + \eta_\lambda)^{d-1}\,\mathrm{Lip}(v_0^*)^2\eta_\lambda \leqslant \lambda C_\lambda,$$

where $C_\lambda = c_d M_\rho \mathcal{R}_Y^3(\mathcal{R}_X + 2\mathcal{R}_Y\lambda)^{d-1}$. Because γ has marginals ρ and μ, the integral of the Fenchel duality gap is equal to the primal gap P :

$$\begin{aligned}\int_{X\times Y} G_{v_0^*}(x,y)\mathrm{d}\gamma &= \int_X v_0^*\mathrm{d}\rho + \int_Y v_0\mathrm{d}\mu - \int_{X\times Y}\langle x|y\rangle\mathrm{d}\gamma \\ &= \int_{X\times Y}\langle x|y\rangle\mathrm{d}\gamma_0 - \int_{X\times Y}\langle x|y\rangle\mathrm{d}\gamma =: P.\end{aligned}$$

Gathering these estimates, we obtain the upper bound

$$\int_{X\times Y} \|y - \nabla v_0^*(x)\|^2\,\mathrm{d}\gamma(x,y) \leqslant \frac{2}{\lambda}P + 2\lambda C_\lambda.$$

Taking $\overline{\lambda} = \sqrt{2P/C_0}$, the upper bound becomes $\sqrt{2C_0P} + 2C_{\overline{\lambda}}\sqrt{2P/C_0}$. To obtain an unconditional upper bound, we must bound $C_{\overline{\lambda}}$. Since the supports X and Y are contained in balls of radius $\mathcal{R}_X$ and $\mathcal{R}_Y$, the inner product $\langle x|y\rangle$ is bounded by $\mathcal{R}_X\mathcal{R}_Y$ in absolute value. Thus, the primal gap P is bounded by $2\mathcal{R}_X\mathcal{R}_Y$. This implies that our choice of λ satisfies:

$$\overline{\lambda} = \sqrt{\frac{2P}{C_0}} \leqslant \sqrt{\frac{4\mathcal{R}_X\mathcal{R}_Y}{C_0}}.$$

Since $\lambda \mapsto C_\lambda$ is non-decreasing, $C_{\overline{\lambda}}$ is indeed uniformly bounded by a constant that only depends on $M_\rho, \mathcal{R}_X$, and $\mathcal{R}_Y$ and d. □

4. Application: convergence of entropic Brenier maps

Let X, Y be compact subsets of $\mathbb{R}^d$, let $\rho \in \mathcal{P}^{\mathrm{ac}}(X)$ be a probability density on X and let $\mu \in \mathcal{P}(Y)$. Given $\varepsilon > 0$, we consider the regularized primal problem

$$\max_{\gamma \in \Gamma(\rho,\mu)} \int_{X\times Y} \langle x|y\rangle \mathrm{d}\gamma - \varepsilon H(\gamma \mid \rho \otimes \mu), \tag{4.16}$$

where $H(\gamma \mid \rho\otimes\mu)$ is the relative entropy. The unique solution is the entropic plan, denoted $\gamma_\varepsilon \in \Gamma(\rho,\mu)$. We denote $\gamma_\varepsilon(y \mid x)$ the conditional distribution of y given x under γ_ε, so that $\gamma_\varepsilon = \int_X \gamma_\varepsilon(\cdot \mid x)\mathrm{d}\rho(x)$. The entropic Brenier map $T_\varepsilon : X \to Y$ is then defined at ρ-almost every point x in X as the centroid of $\gamma_\varepsilon(y \mid x)$:

$$T_\varepsilon(x) = \int_Y y \mathrm{d}\gamma_\varepsilon(y \mid x). \tag{4.17}$$

When $\varepsilon = 0$, γ_0 is the standard optimal transport plan, which is supported on the graph of the Brenier map T_0.

A fine understanding of the convergence of the entropy-regularized problem to the unregularized one as $\varepsilon \to 0$ has been the object of many recent works. While quantitative convergence rates of the plans and maps typically require strong regularity assumptions on the unregularized map, the qualitative behavior is well established. In our setting, because the source measure ρ is absolutely continuous and the cost is $-\langle x|y\rangle$, the unregularized optimal plan γ_0 is unique; consequently, it is a standard result that γ_ε converges weakly to γ_0 unconditionally. To quantify this convergence and to apply Theorem 3.4, we will need to control the primal suboptimality gap:

$$\begin{aligned} P_\varepsilon(\rho,\mu) &= \int_{X\times Y} \frac{1}{2}\|x-y\|^2 \,\mathrm{d}\gamma_\varepsilon(x,y) - \int_{X\times Y} \frac{1}{2}\|x-y\|^2 \,\mathrm{d}\gamma_0(x,y) \\ &= \int_{X\times Y} \langle x|y\rangle \mathrm{d}\gamma_0(x,y) - \int_{X\times Y} \langle x|y\rangle \mathrm{d}\gamma_\varepsilon(x,y). \end{aligned}$$

We quote two results providing bounds on this primal gap $P_\varepsilon(\rho,\mu)$. First, Malamut and Sylvestre [21] give a $O(\varepsilon)$ bound on the primal gap for continuous measures, see also [13, 5]. We state their result in the case of compactly supported measures, though it holds more generally for absolutely continuous measures admitting finite moments of order > 2.

Theorem 4.1 (Theorem 3.7 in [21]). *Let X, Y be compact subsets of $\mathbb{R}^d$, and let $\rho \in \mathcal{P}^{\mathrm{ac}}(X)$ and $\mu \in \mathcal{P}^{\mathrm{ac}}(Y)$ both with finite entropy. Then, the primal gap satisfies $P_\varepsilon(\rho,\mu) = \Theta(\varepsilon)$.*

We also note that in the semi-discrete setting, the rate can be improved to $O(\varepsilon^2)$ thanks to a result of Altschuler, Niles-Weed and Stromme [1].

Theorem 4.2 (Theorem 1.1 in [1]). *Let X, Y be compact subsets of $\mathbb{R}^d$, and let $\rho \in \mathcal{P}^{\mathrm{ac}}(X)$ and $\mu \in \mathcal{P}(Y)$. Assume that the density of ρ is continuous and positive in the interior of its connected support, and that μ is finitely supported. Then, the primal gap satisfies $P_\varepsilon(\rho,\mu) = \Theta(\varepsilon^2)$.*

Gathering these theorems with Theorem 3.4, we therefore obtain:

THEOREM 4.3. *Let $X, Y \subseteq \mathbb{R}^d$ be compact and convex. Assume that $\rho \in \mathcal{P}^{\mathrm{ac}}(X)$ has bounded density and that $\mu \in \mathcal{P}(Y)$. Denote T_0 the Brenier map from ρ to μ, $\gamma_\varepsilon \in \Gamma(\rho, \mu)$ the entropic optimal transport between ρ and μ and T_ε the entropic Brenier map. Then,*

$$\|T_\varepsilon - T_0\|^2_{\mathrm{L}^2(\rho)} \leqslant \int_{X\times Y} \|y - T_0(x)\|^2 \,\mathrm{d}\gamma_\varepsilon(x,y) \leqslant C\sqrt{P_\varepsilon}$$

In particular:

- *Under the assumptions of Theorem 4.1, $\|T_\varepsilon - T_0\|_{\mathrm{L}^2(\rho)} = O(\varepsilon^{1/4})$.*
- *Under the assumptions of Theorem 4.2, $\|T_\varepsilon - T_0\|_{\mathrm{L}^2(\rho)} = O(\varepsilon^{1/2})$.*

Proof. Using Jensen's inequality, we have

$$\begin{aligned}
\int_X \|T_\varepsilon(x) - T_0(x)\|^2 \,\mathrm{d}\rho(x) &= \int_X \left\| \int_Y (y - T_0(x)) \mathrm{d}\gamma_\varepsilon(y \mid x) \right\|^2 \mathrm{d}\rho(x) \\
&\leqslant \int_X \int_Y \|y - \nabla v_0^*(x)\|^2 \,\mathrm{d}\gamma_\varepsilon(y \mid x) \mathrm{d}\rho(x) \\
&= \int_{X\times Y} \|y - \nabla v_0^*(x)\|^2 \,\mathrm{d}\gamma_\varepsilon(x,y).
\end{aligned}$$

We then conclude using Theorem 3.4, Theorem 4.1 and Theorem 4.2. □

Remark 4.1 (Comparison with existing convergence rates). Prior to this work, quantitative $\mathrm{L}^2(\rho)$ convergence rates for the entropic Brenier map $T_\varepsilon \to T_0$ were heavily dependent on the regularity of the solution to the unregularized problem.

At the highest level of regularity — assuming for instance $\mathcal{C}^3$ regularity of the unregularized map T_0 and finite integrated Fisher information along the Wasserstein geodesic — Pooladian and Niles-Weed [24, Corollary 1] established a fast rate of $\|T_\varepsilon - T_0\|_{\mathrm{L}^2(\rho)} = O(\varepsilon)$. At a lower regularity level, but still assuming that the unregularized map T_0 is Lipschitz, Malamut and Sylvestre [21] recovered a slower $O(\varepsilon^{1/2})$ rate by combining their primal gap bound (Theorem 4.1) with the stability framework of Li and Nochetto [20]. Beyond these smooth settings, Divol et al. [12, Section 4] recently established the same $O(\varepsilon^{1/2})$ rate for semi-discrete optimal transport problems. This rate could also be recovered by combining Corollary 2.2 from [9] with Theorem 3.5 from [7].

Compared to these results, Theorem 4.3 makes very little regularity assumptions — assuming only compactness of the supports and an L^∞ bound on the source density. The upper bound only depends on the primal gap P_ε, so that any progress in the estimation of this gap will automatically lead to tighter estimates for the stability of the entropic Brenier map.

Remark 4.2 ((Non-)optimality of the exponent). We do not know whether the exponent $\frac{1}{4}$ in Theorem 4.3 is optimal. However, simple examples even in dimension $d = 1$ show that we cannot expect this exponent to better than $1/2$ in general.

Appendix A. Alternative proof of the lower bound on the Kantorovich-Bregman divergence

We propose a second proof of the lower bound on the Bregman divergence (Theorem 2.2) relying on the proximal operator and the Fenchel duality gap. We will use a few definitions and results from convex analysis.

Definition A.1. Given $u : \mathbb{R}^d \to \mathbb{R}$ convex, we define its *Fenchel duality gap* by

$$G_u(x, y) = u(x) + u^*(y) - \langle x|y\rangle \geqslant 0,$$

where u^* is the convex conjugate. Furthermore, given a convex function g on $\mathbb{R}^d$ and a parameter $\lambda > 0$, we define the proximal operator as

$$\operatorname{prox}_{\lambda g}(x) = \arg\min_z \left(\frac{1}{2\lambda} \|x - z\|^2 + g(z) \right).$$

As the next lemma shows, the Kantorovich-Bregman divergence can be expressed in terms of a Fenchel duality gap.

Lemma A.1. *Let $v_0, v_1 \in \mathcal{C}^0(Y)$, and define $u_i = v_i^*$. Then, the Bregman divergence admits the following representation:*

$$B_\rho(v_0, v_1) = \int_X G_{u_1}(x, \nabla u_0(x)) \mathrm{d}\rho(x).$$

Proof. Using the fact that $\nabla \mathcal{K}_\rho(v_0) = -(\nabla u_0)_\# \rho$, we see that the Bregman-Kantorovich difference can be expressed as

$$B_\rho(v_0, v_1) = \int_X \left[u_1(x) - u_0(x) + v_1(\nabla u_0(x)) - v_0(\nabla u_0(x)) \right] \mathrm{d}\rho(x).$$

Since u_0 is convex and since $u_0 = v_0^*$, one has $u_0(x) = \langle x|\nabla u_0(x)\rangle - v_0(\nabla u_0(x))$ for Lebesgue-almost (and thus ρ-almost) every point x in X. Substituting this into the integral gives:

$$\begin{aligned} B_\rho(v_0, v_1) &= \int_X \left[u_1(x) + v_1(\nabla u_0(x)) - \langle x|\nabla u_0(x)\rangle \right] \mathrm{d}\rho(x) \\ &= \int_X G_{u_1}(x, \nabla u_0(x)) \mathrm{d}\rho(x). \end{aligned}$$

□

The next lemma is stated as Lemma 1.1 in [3]. It will allow us to obtain a lower bound on the Bregman divergence.

Lemma A.2. *Let $u : \mathbb{R}^d \to \mathbb{R}$ be convex and let $\lambda > 0$. Then, for any $x, y \in \mathbb{R}^d$, letting $x_\lambda = \operatorname{prox}_{\lambda u}(x + \lambda y)$ we have*

$$G_u(x, y) \geqslant \frac{1}{\lambda} \|x - x_\lambda\|^2.$$

Proof of Theorem 2.2.(ii). As in the other proof, we assume without loss of generality that $X \subseteq \mathrm{B}(0, \mathcal{R}_X)$ and $Y \subseteq \mathrm{B}(0, \mathcal{R}_Y)$. Let $v_0, v_1 \in \mathcal{C}^0(Y)$ and let $u_0 = v_0^*, u_1 = v_1^*$. For $\lambda > 0$ and a point $x \in X$, define $x_\lambda = \operatorname{prox}_{\lambda u_1}(x + \lambda \nabla u_0(x))$. Using a density argument, it is possible to assume without loss of generality that u_1 is continuously differentiable.

Step 1: We first use the representation of the Kantorovich-Bregman divergence B_ρ given in Lemma A.1:

$$B_\rho(v_0, v_1) = \int_X G_{u_1}(x, \nabla u_0(x)) \mathrm{d}\rho(x).$$

Setting $x_\lambda = \mathrm{prox}_{\lambda u_1}(x + \lambda \nabla u_0(x))$, and applying the inequality of Theorem A.2 into the integral, we get

$$B_\rho(v_0, v_1) \geqslant \frac{1}{\lambda} \int_X \|x - x_\lambda\|^2 \, \mathrm{d}\rho(x).$$

Using the first-order condition for the proximal operator and the differentiability of u_1, we have

$$\frac{x - x_\lambda}{\lambda} + \nabla u_0(x) = \nabla u_1(x_\lambda). \tag{A.18}$$

Substituting this back into the integral inequality gives us

$$B_\rho(v_0, v_1) = \lambda \int_X \|\nabla u_0(x) - \nabla u_1(x_\lambda)\|^2 \, \mathrm{d}\rho(x).$$

Step 2. For any $\lambda > 0$, we establish a lower bound by decomposing the squared norm and applying the inequality $\|A + B\|^2 \geqslant \frac{1}{2}\|A\|^2 - \|B\|^2$:

$$\begin{aligned} B_\rho(v_0, v_1) &\geqslant \lambda \int_X \|\nabla u_0(x) - \nabla u_1(x_\lambda)\|^2 \, \mathrm{d}\rho(x) \\ &= \lambda \int_X \|(\nabla u_0(x) - \nabla u_1(x)) + (\nabla u_1(x) - \nabla u_1(x_\lambda))\|^2 \, \mathrm{d}\rho(x) \\ &\geqslant \frac{\lambda}{2} \int_X \|\nabla u_0(x) - \nabla u_1(x)\|^2 \, \mathrm{d}\rho(x) - \lambda \int_X \|\nabla u_1(x_\lambda) - \nabla u_1(x)\|^2 \, \mathrm{d}\rho(x). \end{aligned}$$

Equation (A.18) and the fact that $\nabla u_0(x)$ and $\nabla u_1(x_\lambda)$ both belong to the convex set Y imply that $\|x_\lambda - x\| \leqslant \eta_\lambda := 2\mathcal{R}_Y \lambda$. Applying Theorem 2.3 with $w = u_1$, $R = \mathcal{R}_X$, and $\eta = \eta_\lambda = 2\mathcal{R}_Y \lambda$, we obtain:

$$B_\rho(v_0, v_1) \geqslant \frac{\lambda}{2} \int_X \|\nabla u_0(x) - \nabla u_1(x)\|^2 \, \mathrm{d}\rho(x) - \lambda \left[c_d M_\rho (\mathcal{R}_X + \eta_\lambda)^{d-1} \mathcal{R}_Y^2 \eta_\lambda \right].$$

We then conclude exactly as in the lower bound on the Jensen divergence. □

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